

6th Grade Accelerated Unit 11 - Part 1 Assessment Guide

General Information

Part 1 of this assessment covers two benchmarks: MA.7.NSO.2.1 and MA.7.NSO.2.2. These benchmarks are interconnected. MA.7.NSO.2.2 addresses fluency of operations with rational numbers. Students have been building their fluency with operations throughout the course – first in Units 1 and 2 with positive rational number, then in Unit 5 with integers, and now in Unit 11 with positive and negative rational numbers.

According to the course description, the expectation of grade 6 accelerated students is that they are procedurally fluent with performing all four operations with rational numbers, including negative values by the end of the course.

Procedural fluency of rational numbers is one of the five areas of emphasis of learning in grade 6 accelerated. Students will reach this fluency at different times and at different rates. Teachers are encouraged to continue to support student fluency throughout the year by utilizing instructional routines that help students grow in their flexibility of recognizing equivalent forms of rational numbers and performing operations with rational numbers.

The National Council of Teachers of Mathematics (NCTM) has the following position on procedural fluency.

Procedural fluency is a critical component of mathematical proficiency. Procedural fluency is the ability to apply procedures accurately, efficiently, and flexibly; to transfer procedures to different problems and contexts; to build or modify procedures from other procedures; and to recognize when one strategy or procedure is more appropriate to apply than another. To develop procedural fluency, students need experience in integrating concepts and procedures and building on familiar procedures as they create their own informal strategies and procedures. Students need opportunities to justify both informal strategies and commonly used procedures mathematically, to support and justify their choices of appropriate procedures, and to strengthen their understanding and skill through distributed practice.

Please note that NCTM did not state that fluency involves speed, but it did state that it involves efficiency. Efficiency can look different for each individual student. Student confidence can be built by having students share different strategies by using instructional routines such as “or you could,” where students share their different strategies or equivalent results. Another instructional routine that can be used is “why not,” where students examine a strategy that is

	not the most efficient and explain why the strategy is not a good fit. These and other instructional routines help students build their mathematical agency and flexibility. Teachers are encouraged to use this assessment to determine which students have reached procedural fluency on all four operations with rational numbers and which students need further support on this procedural fluency. It is suggested that teachers look at student work to determine whether fluency is being hampered by misconceptions or incorrect methods. This assessment guide includes different possible methods of determining the value of the given expression. Students may use other methods that are not shown. The guide purposefully shows these methods so the teacher can consider alternatives. Students apply their knowledge of performing operations with rational numbers within the benchmark MA.7.NSO.2.1. Teachers are encouraged to evaluate student work to determine whether student mistakes are misunderstanding what operations should be applied first or if the students do not understand absolute value.
Calculator Information	The questions on this assessment are no calculator.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1	Benchmark(s)	MA.7.NSO.2.2					
		Recommended Scoring (8 Total Points)	Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception. All equivalent answers should be expected. The possible methods shown are likely the most common methods that students use. But students who choose other methods and correctly determine an equivalent value should not lose points.					
			Additional Information This question is assessing student flexibility with different forms of rational numbers. Teachers should look for student understanding of equivalency.					
<table><tr><th>Using Fractions</th><th>Using Decimals</th></tr><tr><td> $6\frac{3}{5} + 2.38$ <table><tr><td>$2.38 = 2\frac{38}{100}$$6\frac{3}{5} = 6\frac{60}{100}$$6\frac{60}{100} + 2\frac{38}{100}$$8\frac{98}{100} \text{ or } 8\frac{49}{50}$</td><td>$2.38 = 2\frac{38}{100}$$2\frac{38}{100} = 2\frac{19}{50}$$6\frac{3}{5} = 6\frac{30}{50}$$6\frac{30}{50} + 2\frac{19}{50}$$8\frac{49}{50}$</td></tr></table> </td><td> $6\frac{3}{5} + 2.38$ $6\frac{3}{5} = 6\frac{6}{10} = 6.6$ $6.6 + 2.38$ 8.98 </td></tr></table>		Using Fractions	Using Decimals	$6\frac{3}{5} + 2.38$ <table><tr><td>$2.38 = 2\frac{38}{100}$$6\frac{3}{5} = 6\frac{60}{100}$$6\frac{60}{100} + 2\frac{38}{100}$$8\frac{98}{100} \text{ or } 8\frac{49}{50}$</td><td>$2.38 = 2\frac{38}{100}$$2\frac{38}{100} = 2\frac{19}{50}$$6\frac{3}{5} = 6\frac{30}{50}$$6\frac{30}{50} + 2\frac{19}{50}$$8\frac{49}{50}$</td></tr></table>	$2.38 = 2\frac{38}{100}$ $6\frac{3}{5} = 6\frac{60}{100}$ $6\frac{60}{100} + 2\frac{38}{100}$ $8\frac{98}{100} \text{ or } 8\frac{49}{50}$	$2.38 = 2\frac{38}{100}$ $2\frac{38}{100} = 2\frac{19}{50}$ $6\frac{3}{5} = 6\frac{30}{50}$ $6\frac{30}{50} + 2\frac{19}{50}$ $8\frac{49}{50}$	$6\frac{3}{5} + 2.38$ $6\frac{3}{5} = 6\frac{6}{10} = 6.6$ $6.6 + 2.38$ 8.98	
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Question	2	Benchmark(s)	MA.7.NSO.2.2
		Recommended Scoring (8 Total Points)	Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception. All equivalent answers should be expected. The possible methods shown are likely the most common methods that students use. But students who choose other methods and correctly determine an equivalent value should not lose points.
Using $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ $-4\frac{5}{24} - \frac{21}{16}$$\frac{-101}{24} + \frac{-21}{16}$$\frac{-101 \cdot 16 + -21 \cdot 24}{24 \cdot 16}$$\frac{-1616 + -504}{384}$$\frac{-2120}{384}$	Using a common denominator of 48 $-4\frac{5}{24} - \frac{21}{16}$$-4\frac{5}{24} = -4\frac{10}{48}$$\frac{21}{16} = 1\frac{5}{16}$$1\frac{5}{16} = 1\frac{15}{48}$$-4\frac{10}{48} - 1\frac{15}{48}$$-5\frac{25}{48}$	Additional Information	Students who use the method, $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$, may not “reduce” to lowest terms. The answer as shown is sufficient. Simplifying fractions to lowest terms is not learned until grade 6, MA.6.NSO.3.1, and is taught as finding common factors of the numerator and denominator. Students who use a common denominator may not use 48 as the common denominator.

Question	3	Benchmark(s)	MA.7.NSO.2.2
		Recommended Scoring (8 Total Points)	Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception. All equivalent answers should be expected. The possible methods shown are likely the most common methods that students use. But students who choose other methods and correctly determine an equivalent value should not lose points.
Using Decimals	Using Fractions		
$-4.63 \cdot 3.9$ $\begin{array}{r} 4.63 \\ \times 3.9 \\ \hline 4167 \\ 13890 \\ \hline 18.057 \end{array}$ -18.057	$-4.63 \cdot 3.9$ $-4\frac{63}{100} \cdot 3\frac{9}{10}$ $\frac{-463}{100} \cdot \frac{39}{10} = \frac{-18057}{1000}$ $\frac{-18057}{1000}$	Additional Information	Two possible methods are shown.

Question	4	Benchmark(s)	MA.7.NSO.2.2
		Recommended Scoring (8 Total Points)	Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.
			The two methods shown are representative of how flexible the benchmark allows students to be. Teachers often ask about the expectations of statewide assessments. The draft test item specifications do not require that answers be in the most simplified form so students should be able to answer equation editor questions without simplification. Multiple choice questions may require students to recognize equivalent responses which means students are applying their knowledge from MA.7.NSO.1.2. Since this is requiring knowledge of two different benchmarks it is not likely students would have to simplify when a question is assessing their knowledge of MA.7.NSO.2.2 because Florida's assessments assess one benchmark for each question.
Using Long Division	Using Fractions	Additional Information	
$32.5892 \div (-7.91)$ $\begin{array}{r} 7.91 \overline{) 32.5892} \\ 791 \overline{) 3258.92} \\ \underline{4.12} \\ 791 \overline{) 3258.92} \\ \underline{-3164} \\ 949 \\ \underline{-791} \\ 1582 \\ \underline{-1582} \\ 0 \end{array}$ -4.12	$32.5892 \div (-7.91)$ $32 \frac{5892}{10000} \div -7 \frac{91}{100}$ $\frac{325892}{10000} \div \frac{-791}{100}$ $\frac{325892}{10000} \cdot \frac{100}{-791}$ $\frac{32589200}{-7910000}$ $-\frac{32589200}{7910000} \text{ or } -\frac{325892}{79100} \text{ or } -\frac{412}{100}$ or any other equivalent		

Question		5	Benchmark(s)	MA.7.NSO.2.2
Using Fractions $2.75 \cdot \frac{19}{8}$ $2\frac{3}{4} \cdot \frac{19}{8}$ $\frac{11}{4} \cdot \frac{19}{8}$ $\frac{209}{32} \text{ or any other equivalent}$		Using Decimals $2.75 \cdot \frac{19}{8}$ $\frac{19}{8} = 2\frac{3}{8}$ $\frac{3}{8} = 0.375$ $2.75 \cdot 2.375$ $\begin{array}{r} 2.375 \\ \times 2.75 \\ \hline 11875 \\ 166250 \\ 475000 \\ \hline 6.53125 \end{array}$	Recommended Scoring (8 Total Points)	Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.
			Additional Information	This question is assessing student flexibility with different forms of rational numbers. Teachers should look for student understanding of equivalency.

Question	6	Benchmark(s)	MA.7.NSO.2.2
Using Fractions $8\frac{2}{3} \div \left(-\frac{104}{33}\right)$ $\frac{26}{3} \div \left(-\frac{104}{33}\right)$ $\frac{26}{3} \cdot \left(-\frac{33}{104}\right)$ $\frac{\overset{1}{\cancel{26}}}{\cancel{3}} \cdot -\frac{\overset{11}{\cancel{33}}}{\cancel{104}}$ $\frac{1}{1} \cdot -\frac{11}{4}$ $-\frac{11}{4} \text{ or any equivalent}$	Using Decimals $8\frac{2}{3} \div \left(-\frac{104}{33}\right)$ $8.\overline{6} \div -3.\overline{15}$ $3.\overline{15} \overline{)8.66666}$ $\begin{array}{r} 2.75 \\ 315 \overline{)866.666} \\ \underline{-630} \\ 2366 \\ \underline{-2205} \\ 1616666\ldots \\ \underline{-1616666\ldots} \end{array}$	Recommended Scoring (8 Total Points)	Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.
		Additional Information	The key shows students using common factors to “simplify” the answer. Students are not required to do this.

Question	7	Benchmark(s)	MA.7.NSO.2.2
0.0558		Recommended Scoring (8 Total Points)	Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.
		Additional Information	Watch for student misconceptions based on place value. Students are not likely to determine the sum using fractions, but they should have the freedom to do so if they wish.

Question		8	Benchmark(s)	MA.7.NSO.2.2												
<table><thead><tr><th>Fractions</th><th>Decimals</th></tr></thead><tbody><tr><td>$-\frac{34}{5} \cdot 1\frac{1}{10}$</td><td>$-\frac{34}{5} \cdot 1\frac{1}{10}$</td></tr><tr><td>$-\frac{34}{5} \cdot \frac{11}{10}$</td><td>$-6\frac{4}{5} \cdot 1\frac{1}{10}$</td></tr><tr><td>$-\frac{374}{50}$</td><td>$-6\frac{8}{10} \cdot 1\frac{1}{10}$</td></tr><tr><td></td><td>$-6.8 \cdot 1.1$</td></tr><tr><td></td><td>-7.48</td></tr></tbody></table>			Fractions	Decimals	$-\frac{34}{5} \cdot 1\frac{1}{10}$	$-\frac{34}{5} \cdot 1\frac{1}{10}$	$-\frac{34}{5} \cdot \frac{11}{10}$	$-6\frac{4}{5} \cdot 1\frac{1}{10}$	$-\frac{374}{50}$	$-6\frac{8}{10} \cdot 1\frac{1}{10}$		$-6.8 \cdot 1.1$		-7.48	Recommended Scoring (8 Total Points)	Consider following student work to determine if a mistake is a minor miscalculation, a major error, or a misconception.
			Fractions	Decimals												
$-\frac{34}{5} \cdot 1\frac{1}{10}$	$-\frac{34}{5} \cdot 1\frac{1}{10}$															
$-\frac{34}{5} \cdot \frac{11}{10}$	$-6\frac{4}{5} \cdot 1\frac{1}{10}$															
$-\frac{374}{50}$	$-6\frac{8}{10} \cdot 1\frac{1}{10}$															
	$-6.8 \cdot 1.1$															
	-7.48															
			Additional Information	This question is an excellent example where students with flexible thinking may find using decimals are more efficient than fractions. Using a similar question for student discussion may help to build student fluency.												
Question		9	Benchmark(s)	MA.7.NSO.2.1												
☐ $8 \cdot (-6)^6$ ☒ $8 \cdot 3(64)$ ☒ $24 \cdot (-2)^6$ ☐ $8 \cdot 3(-12)$ ☒ $2^3 \cdot (-2)^6 \cdot 3$			Recommended Scoring (12 Total Points)	Consider 4 points for each correct answer. Consider deducting 2 points for each incorrect answer. Consider no points if all are selected.												

Question	3	Benchmark(s)	MA.7.NSO.2.1
-3.57		Recommended Scoring (6 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	10	Benchmark(s)	MA.7.NSO.2.1
$\frac{38}{7}$		Recommended Scoring (6 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	15	Benchmark(s)	MA.7.NSO.2.1
☐ -11 ☒ 0 ☐ $\frac{64}{-8} + \frac{6}{-2}$ ☒ $\frac{64}{16} + \frac{8}{-2}$ ☐ $8^2 \div \left[(-4)^2 + \left(-\frac{1}{2} \right) \right] (2^3)$ ☐ $\left[8^2 \div (-4)^2 + \left(-\frac{1}{2} \right) \right] (2^3)$ ☒ $(8^2 \div (-4)^2) + \left(\left(-\frac{1}{2} \right) (2^3) \right)$		Recommended Scoring (12 Total Points)	Consider 4 points for each correct answer. Consider deducting 2 points for each incorrect answer. Consider no points if all are selected.